



Brazilianite

△ Pale yellow brazilianite rough

Brazilianite was only discovered and named – after the South American country of its discovery – in 1945, making it a relatively “new” gemstone. Most brazilianite is chartreuse-yellow to pale yellow in colour, and it is relatively hard for a phosphate mineral; it is also brittle. It is scarce, and very little gem-grade material is found each year. Because of these factors, it is faceted only for collectors. Since brazilianite was recognized, small amounts of the gem have also been found in Maine and New Hampshire in the USA.

Specification

Chemical name	Sodium aluminium phosphate	Formula	$\text{NaAl}_3(\text{PO}_4)_2(\text{OH})_4$
Colours	Yellow, green	Structure	Monoclinic
Hardness	5.5	SG	3.0
RI	1.60–1.62	Lustre	Vitreous
Streak	Colourless	Locations	Brazil; Maine and New Hampshire, USA

Prismatic crystals | Rough | This group of finely formed, prismatic brazilianite crystals with accessory apatite originates from Minas Gerais, Brazil.



Good transparency



Crystal | Rough | This strikingly coloured lime green brazilianite crystal comes from the Brazilian state of Minas Gerais. It is part of the Smithsonian's gem and mineral collection.

Facets on back of stone are visible



Emerald cut | Cut | For this faceted gem, the cutter has chosen a step-cut variety – the emerald cut – to emphasize the stone's fine colour and transparency.

Cutting brazilianite

Handled with care

Vivid yellow brazilianite's beautiful appearance would make it a popular gem, were it not for two factors – its fragility and its brittle texture. To cut it, the lapidary (gem-cutter) usually sticks the stone to its holder with easily removable adhesive, and must take great care not to knock it against anything. Likewise, any vibration in the grinding and polishing stages will shatter the brittle stone. The high level of skill required means that faceted stones are comparatively rare.



Gem cutting Brazilianite requires extreme care in cutting, or its brittleness will cause it to shatter.

Star facet



Fancy cut | Cut | Featuring a classic example of a fancy cut, this yellow brazilianite gemstone has been faceted in a traditional triangular step cut.

Table facet



Brilliant cut | Cut | This greenish brazilianite is faceted in a standard brilliant cut with 52 facets, as opposed to the 58-facet version, which is more common.